Contents

1 Introduction

1.1 Background

With the expansion of the unmanned vehicle (land, air, surface and underwater) in many battlefield and disaster scenarios, there is a demand for methods of controlling these. Whilst autonomy is garnering a lot of attention in this way, there is a need for the expertise and skill that a human controller brings, whilst standing them off from the dangers of hazardous environments.

An area of great interest currently is in unmanned air vehicles, where development is being taken up in earnest by organisations from major engineering firms to undergraduate project groups. Another field which is developing rapidly, however, is that of marine autonomy. This is evidenced by the speech given by the First Sea Lord at the DSEI conference, where the need for a "fleet of crewed, uncrewed, and autonomous platforms"[1] was identified. Additionally in the merchant naval space, the uncrewed platforms are being recognised, with the Workboat Code Edition 3 adding Annexe 2 in 2025 to specifically target what it terms "Remotely Operated Unmanned Vessels (ROUVs)" in section 1.1.1 [2].

Companies such as Saab AB, Thales Group, and L3Harris [3][4][5] have been making progress towards realising the vision of a "hybrid fleet" whilst companies such as ACUA Ocean, Deep Ocean Engineering, and HD Hyundai [6][7][8] are investing in ROUVs for the commercial market. Additonally, the US Navy has begun fielding ROUVs, for example the Fleet class Unmanned Surface Vessel built by Textron [9]. In all of these cases autonomy has been the end goal, to perform dangerous task such as mine countermeasures, tedious ones such as survey work [10], and other routine but challenging tasks such as docking [11].

Autonomy in surface ships is a significant challenge, however. Any oceangoing vessel must comply with national and international law, which not only set requirements for the vessels but also for the behaviours that they must exhibit. A key example of this is the Convention on the International Regulations for Preventing Collisions at Sea (COLREGs) [12]. COLREGs set out rules determining the lights, shapes, and sounds that a vessel must display depending on its state, but also they describe responsibilities of different vessels in different situations. These rules are often subjective, however, and were designed for human navigators [11]. Implementing COLREGs into autonomous systems is as such an area of active development.

To operate in national waters, a ship must be approved by a national agency, which in the UK is the Maritime and Coastguard Agency (MCA). To do this, the vessel must show that it is compliant with Workboat Code Edition 3 (WBC3). ROUVs are now goverened specifically by Annexe 2, which sets out requirements on the ROUVs and their control stations. The control stations on a ROUV can be varied, from a wheelhouse on a retrofit vessel to a land based operations centre. Of note is that ROUVs through WBC3 must comply with a number of standards [13], including BS EN 61508: 2010 [14]. This standards concerns the functional safety of electrical/electronic/programmable electronic safety related systems, such as the emergency stop system on a vessel or its damage control systems. BS EN 61508 defines Safety Integrity Levels (SIL), which correspond to a number of dangerous failurs per hour. For example, achieving SIL 1 compliance requires a failure rate no greater than 10^{-5} dangerous failures per hour, whilst SIL 3 requires no greater than 10^{-9}. Whilst implementations of BS EN 61508 are common in many sectors (for example, ISO 26262 for the automotive industry [15]), ensuring that any automated controllers are also compliant is a challenge. To ensure that an ROUV has a safety case sufficient to be granted approval from the MCA, remote crew operation capability is often required.

1.2 Problem Statement

As mentioned above, autonomy in workboats is as such becoming increasingly mature and is being deployed to many different platforms. Until a level of assurance in the autonomous controller's capabilies and reliability is reached (if such a level will ever be reached), human oversight of the vessel will be required in some from. Even in situations where autonomy is not the end goal for the vessel, there is a need for a method of removing the crew from the vessel in hazardous conditions. This can reduce the chance and/or severity of accidents (man overboard, injuries from machinery, loss of platform) whilst potentially improving quality of life for those crews (e.g. protection from the elements). Moreover, there is an opportunity for workboats to be constructed without a wheelhouse and on board controls, which could reduce the cost, complexity, and displacement of the platform. In all of these situations remote control of the vessel is desirable: to keep watch over autonomy or remove the crew from the vessel. As mentioned above, Workboat Code edition 3 Annexe 2 [2] prescribes requirements for an ROUV, but does not prescribe solutions to this issue.

Existing control stations for ROUVs are large and bulky. They often are entire rooms, with multiple monitors, a PC like interface, and additional hardware controls set up with communications software built in to enable control or keeping watch of the ROUV beyond visual range. In the case where they are designed for use outside (such as in the case of line of sight control or), they are still too large and heavy to be carried by an operator, needing instead to be set up from a single vantage point (such as the Deep Trekker BRIDGE Console [16] and MarineNav Oceanus [17]). This is ideal for situations that are beyond visual range, such as mission operation, however within line of sight this solution becomes significantly less desirable, as being able to walk around a harbour or be aboard a vessel whilst controlling it may provide better situational awareness. A possible solution is the SRoC Series of ground control station provided by UXV Technologies [18]. It is a control tablet that is man portable, providing control of a remote vehicle over radio communications using hardware buttons and joysticks built into the tablet. This solution is however restricted in its platform agnostic design, failing to offer specific needs of a marine craft (such as throttle control) and integrates with a proprietary communications suite.

As such, there is a need of a bespoke, marine centred remote control unit (RCU) which is man portable and capable of providing remote control of an ROUV within line of sight.

1.3 System Narrative

In designing an RCU, the manner in which the operator issues commands and receives information is important. The RCU will have to be provide situational awareness that is at least as good as a crew member would have aboard the vessel. It also must be able to provide the same functionality and level of control as those controls normally found aboard the platform. Additionally, it must be capable of connecting to the vessel in a reliable and secure way. Such a communication channel must be capable of providing an appropriate SIL, and must permit the transmission of commands, information, and alerts simultaneously. There a number of approaches to this that are dependant on the setting in which the controller would be used.

There are also specific needs that an RCU will need to meet introduced by the remote control. The platform can only accept one controller at a time to avoid complexity in controller hierarchy. As such, the RCU will need to manage assuming control of the vessel, both from a "cold start" and from an exant controller. Additionally, the RCU will need to manage different methods of control behaviour (open water operation, confined spaces, and dynamic station keeping) whilst also being able to manage a degraded communication environment (such as in inclement weather).

Managing ROUV subsystems is also core functionality that must be met. Control of the propulsion and power systems is necessary for controlling movement and for enabling a "cold start" scenario. COLREGs 1972 mandates shapes, lights, and sounds that must be managed depending on the boat's situation and movements, and must be controlled as well [12]. Additionally, the emergency system aboard the vessel must be controlled as well. This includes both alerts to the operator and the emergency stop functionalities. To that end, the vessel must also comply with all relevent legislation, and the RCU must enable the vessel to meet those requirements [2] [12].

1.4 Aims and Objectives

This report aims to provide an initial architectural solution to a man portable RCU that can control an ROUV within line of sight in compliance with all relevant legislation, which can be then used to inform the design of a realised solution.

To do this, this report has the following objectives:

1. To identify scenarios, capabilites, and use cases for an RCU, whilst understanding the relationships between different use cases and the actors external to the system.

2. To define a set of user activities that the RCU would need to be capable of facilitating derived from the scnearios and use cases.

3. To derive a set of requirements from the activities, use cases, and relevant legislation, and to functionally decompose them to inform the structure of the RCU.

4. To identify the constituent parts of a solution, defining their properties and internal structures, allocating them to the functional requirements, and structuring them such that the RCU would be able to facilitate the user activities and scenarios.
5. To supplement the behavioural model of the architected solution with respect to the structral model, to further understand the interaction between the actors and subsystems.

2 Technical Results

2.1 Functional Requirements Definition

A set of functional requirements has been defined and captured in the architectural model. These requirements were elicited from the problem statement and the system narrative, along with key legislation. This set comprises four high-level requirements (see figure), pertaining to four key functionalities that the RCU must enable:

1. Taking control of the ROUV
2. Operating the ROUV
3. Displaying information about the ROUV and its surroundings
4. Enabling the control of the ROUV's subsystems

This was captured in a SysML 1.5 [19] requirements diagram using Sparx System's Enterprise Architect [20]. A requirements diagram was used over traditional requirements management techniques (such as document based approaches) as it allows for a diagrammatic and logical requirement structure to be captured in the model itself. It also allows for other model elements to be related to requirements, such that traceability between the requirements and elements can be communicated. Key to this approach is the containment relationship, which as shown in Section 2.2 is further used to define the functional trees that requirements belong to.

When writing the requirements set, the system boundary was set at the human machine interface (HMI) controls and at the connectivity subsystem. This leaves the operator and the ROUV itself as external actors on the system. As such, requirements such as for training the operator or for the ROUV's handling of RCU data is not considered. For this stage in solution architecture, the two key sources of requirements beyond the problem statement were WBC3 Annex 2 [2] and COLREGs [12]. This generally concerns the management of the ROUV subsystems and the communication of information and alerts to the operator. Other legislature and standards (such as those set out by OFCOM for radio communications) have been omitted, so that complexity is managed. Additionally, the requirements all use the `«functionalRequirement»` stereotype. This is defined in Annex E of the SysML 1.5 specification, and is implemented as standard by Enterprise Architect. This adds granularity to the requirements set in separating the functional requirements from any non-functional requirements that may be added at a later point. To this end, other requirements (interface, performance, physical, and design requirements) have not yet been elicited in this work. These were unnecessary for the initial architecture to be drafted, however would need to be added to complete the set and ensure that all consideration are accounted for in design.

Requirements were also drafted from the creation of scenarios. The main flow scenario was shore based operation of the ROUV prior to any open water work. This included operating in harbour, transiting to the area of operations, and recovery post-mission. An alternative scenario of rescuing the ROUV after a loss of communications/other malfunction leaves it stranded was imagined as an example of alternative behaviour that the architecture will need to be able to accommodate. Both of these scenarios define the need to operate the ROUV, both in open water and in harbour. Additionally, common to these scenarios is the need to take control of the ROUV. The ROUV will have a controller that is "in control," such as an autonomy system or coxswain. For the RCU to control the ROUV, it will need a method of taking control. This was deemed sufficiently distinct from the other three top level requirements and was set at this level.

2.2 Functional Decomposition, Elaboration, and Allocation

Following the definition of the top level requirements, these were functionally decomposed until atomic requirements were reached. These decomposed requirements again use the containment relationship to show ownership by their parent, defining functional trees as mentioned above. The functional trees do not exist in isolation of each other, however, and relations between different requirements has been captured. For example, requirement 3.1 (display navigational data) is owned by requirement 3 (Provide Information) as it is a information requirement, but is derived from requirement 2 (Control ROUV Navigation). In order to control the ROUV, it is necessary to understand its current movement and behaviour (see figure).

The requirements set in this state is not complete; level 1 and 2 requirements can be decomposed further, and no consideration has been given to subsystem requirements. They are in some cases, however, elaborated upon using the «refines» relationship from use cases, as discussed in Section 2.4. This provides further traceability through to the behavioural model of the system, and specifically the capabilities that stakeholders require of the system.

Allocation of system requirements to structural parts is done through the«satisfies» relation (see figure). The blocks that have been allocated to in this way were earmarked for detailed definition, and traceability is maintained throughout the model. This avoids the situation where the system does not fully meet its requirement set, and can enable verification of the system to be focussed on the subsystem that is responsible for each requirement. In this work, some subsystems are included that have not had any requirements allocated to them; this is the result of incomplete requirements. Whilst in a complete requirement set this would indicate that the subsystem is superfluous, at this initial stage of architecture this is not the case.

This requirements set captures the core functionality of the RCU that can then be taken through to structural definition.

2.3 System Structure Definition

Following the definition and decomposition of requirements, a structural model of the RCU was created. This was done using a SysML block definition diagram, along with internal block diagrams created as children of some blocks. The blocks are all defined in a hierarchical structure, with a common parent block called RCU. The system was designed following a hub and spoke structure, with a central "processor" sitting between the other parts. All parts were related to the RCU block using parts reference relationships.

Of particular focus in this work was the "HMI Components" block, which is allocated to a number or requirements. This itself has been decomposed into four subsystems and has had its internal structure defined in an IBD. These subsystems have then had component parts defined which are then used as to type full ports (see below). The communications subsystem, which sits at the boundary between the RCU and the ROUV, has also been modelled, though to less detail than the HMI components. For completeness, the chassis and power subsystem have been included as placeholders. The processor in this stage of the architecture currently serves to manage the connections between the different subsystems in the RCU. This decision has been chosen both to manage complexity and to inform future system design, mimicking the architecture of a PC or a mobile device. In this way the processor can be seen as a "translation layer" between the HMI components and the connectivity subsystem.

Internally, the HMI Components have been modelled as passing items between the RCU Operator and the processor. The Components are varied and provide a different types of inputs, which have been typed as "humanInputs." Each part takes these inputs and converts them into "controlVoltages," which are common can be passed onto the processor. Similarly, the display and headset take data from the processor and output it to the RCU operator. The value type "data" is a generalisation of controlVoltages, image, and sound, so that the proxy ports connecting the HMI components to the processors can be of the same type, again to manage complexity.

The RCU IBD defines the rest of the system's internal behaviour. Two generic ports have been defined on the system boundary as analogues for the RCU operator and the ROUV controller, which is not within the boundary. This keeps the architecture platform agnostic, with any integration to an actual platform requiring definition of this interface. This diagram shows the translation layer structure of the processor, with value properties typed as data (or its specialisations) representing the item flow at an instance in time. This is then connected to the connectivity subsystem, whose components define the ports. In this, the distinction between alerts and information has been modelled, as is required by WBC3[2]. Moreover, the neccesity of two-way communication has been modelled through the bi-directional item flow to "duplexLinkToROUV", as WBC3 mandates certain information being displayed at all times, including when the RCU is sending commands. As mentioned in Section 1.3, the method of communication is important.

2.4 System Interface Behaviour

The behavioural model of the system has also been modelled, with the main behaviour focussing on the normal operation scenario. Initially, a SysML activity hierarchy [19] was created, within which are a number of activity diagrams. These activity diagrams are in some cases comprised of activities themselves. These are linked by part references to the Normal Operation top level activity, which itself has no diagram. Due to the HMI nature of the RCU, all diagrams share three partitions: RCU Operator, RCU, and ROUV. In almost all cases, the RCU

Operator and ROUV never interact directly; the RCU is the sole means of communication between the two. This is not true specifically in the "Hand Over Control" activity defined in "Prepare ROUV for boarding" (whilst the parent activity is not part of Normal Operation, the child activity is). This is because in this specific situation, the ROUV is being handed off to crew aboard the vessel, where the ROUV could reasonably be taken "offline." This is as such a generalisation of the handover relationship. The SendSignal and ReceiveEvent kinds of action have been used to represent interrupts sent between the RCU and the ROUV. This is distinct from the atomic actions, which activate solely from tokens passed along control flows or items along object flows (i.e. synchronous events). Another feature of note is the use of «continuous» object flows in the Take Control activity diagram. This represents the continuous, broadcast nature of the ROUV's information channels. This is to enable any connected RCU to see the control positions and other information at all times (some of which is mandated by [2]). This does pose a security concern in that the information, if constantly broadcast, could be intercepted by a threat actor. To mitigate this, a method of encryption should be employed for critical information, though this is without the scope of this work at this stage.

Following the definition of the RCU structure, the system operation calls were modelled through SysML sequence diagrams [19]. These have been produced for two scenarios: initial RCU connection (in RCU connection SD), and RCU General Sequences. RCU connection SD is an example of the subsystem-level sequence modelling that will be employed over the rest of the possible procedures and scenarios. It again shows the intermediary nature of the processor, and also it's status as a system controller (in owning the initialiseRCU method). The RCU General Sequences diagram shows.

2.5 Alternative Behaviours

2.6 Architecture Analysis

There is a risk that the communication link between the ROUV and the RCU would require a radio frequency that is not easily accessible. Most frequency bands are managed by OFCOM in the UK and require a license to use. Acquiring a license for a different frequency band than has already been acquired for a vessel introduces cost and integration delays that could be avoided with the use of a frequency band without the need of a license. To check this, a simplified model of the harbour environment was used. Free Space Path Loss (FSPL) can be calculated as follows: $FSPL = 20\log_{10} d + 20\log_{10} f + 32.44 - G_t - G_r$, where d is the distance of measurement in kilometres, f is the frequency of the transmission in megahertz, and G_t and G_r are the antenna gains of the transmitter and receiver, respectively [21][22]. For the receiver to receive valid commands reliably, the FSPL must be less than the receiver sensitivity value. Assuming that the communication is over LDP433 (A frequency band at 433 MHz exempt from licensing under OFCOM's IR 2030[23]), the equation can be solved for d.

$$
\begin{aligned}
\Rightarrow 20\log_{10} d =& FSPL + G_t + G_r - 32.44 - 20\log_{10} f \\
\Rightarrow \log_{10} d =& \frac{(FSPL + G_t + G_r)}{20} - 1.622 - \log_{10} f \\
\Rightarrow d =& 10^{\left(\frac{(FSPL+G_t+G_r)}{20} - 1.622 - \log_{10} f\right)} \\
let f \equiv& 433 MHz \\
G_t \equiv& G_r \equiv 1.5dB \\
FSPL \equiv& 100dB \\
\Rightarrow d =& 10^{\left(\frac{(100+1.5+1.5)}{20} - 1.622 - \log_{10} 433\right)} \\
\Rightarrow d =& 10^{(5.15-1.622-2.636)} = 10^{0.892} \\
\Rightarrow d \approx& 7.8km
\end{aligned}
$$

Taking an acceptable line of sight range of 5 km, it is safe to assume that LDP433 is a suitable frequency band and can be used on the RCU, enabling this method of communication over another (such as fibreoptic link).

3 Conclusion

3.1 System Specification

3.2 Achievement of Aims and Objectives

3.3 Highlights and Recommendations

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